

Requirements

Python Version

Python 3.8 or higher is required to run this tool.

Installing Dependencies

Run the following command to install all required packages:

```
pip install numpy pandas scikit-learn scipy matplotlib
```

Package versions tested:

- numpy >= 2.4.1
- pandas >= 3.0.0
- scikit-learn >= 1.8.0
- scipy >= 1.17.0
- matplotlib >= 3.10.8

```
Command Prompt
Microsoft Windows [Version 10.0.26200.8246]
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C:\Users\Aasfi>python --version
Python 3.14.2

C:\Users\Aasfi>python -c "import numpy; print(numpy.__version__)"
2.4.1

C:\Users\Aasfi>python -c "import pandas; print(pandas.__version__)"
3.0.0

C:\Users\Aasfi>python -c "import sklearn; print(sklearn.__version__)"
1.8.0

C:\Users\Aasfi>python -c "import scipy; print(scipy.__version__)"
1.17.0

C:\Users\Aasfi>python -c "import matplotlib; print(matplotlib.__version__)"
3.10.8

C:\Users\Aasfi>
```

Data

Download the 5 CSV files from the official lab repository or my github repository:

<https://github.com/ideas-labo/ISE/tree/main/lab1> or

<https://github.com/aasfimanwar/ise-coursework> (data folder)

Note: I have changed the file name from incubator-mxnet.csv to mxnet.csv in my repository

Place the following 5 CSV files into a folder called data/ in the same directory as classify.py:

- TensorFlow.csv
- PyTorch.csv
- Keras.csv
- Mxnet.csv
- Caffe.csv

Tested Environment

- OS: Windows 11
- Python: 3.14.2